

An Intuitive Review of Linear Regression

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Abstract

One of the most fundamental techniques in Machine Learning and statistics is Linear Regression. It is used to model the relationship between one or more independent variables and dependent variable. The method assumes a linear relationship and estimates the best-fitting line that minimizes prediction error which is mostly called Mean Squared Error (MSE). The Idea of how this model learn comes from Gradient descent. It is one of the most widely used optimization algorithms in machine learning. It is designed to minimize cost functions by iteratively updating model parameters in the direction of the steepest descent of the gradient.

1 Introduction

Linear regression is simple, but it is one of the most powerful methods in machine learning and statistics. It is widely used to model the relationship between independent variables and a dependent variable, providing a straightforward yet effective way to make predictions and analyze data trends. Despite its simplicity, linear regression serves as the foundation for many advanced models.

The Idea of Gradient Descent comes from minimizing cost function, which measures the distance between actual value (y) and predicted value ($\hat{y}$). This is how it makes it possible for model to learn.

2 The Idea of Linear Regression

The core idea behind Linear Regression is to approximate the relationship between input variables and the output using a straight line. This line is defined by the following equation.

$$\hat{y} = wx + b$$

where:

- $\hat{y}$: Model prediction
- w : Weight (slope) of the line
- x : Input feature
- b : Bias (y-intercept)

In real world examples we see multiple features and resulting in generalized form as shown below.

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b \tag{1}$$

The challenge is to determine the best parameters of 'w' and 'b' such that the model's predictions are as close as possible to the actual observations. To achieve this, we measure the error between predicted and true values, commonly using the Mean Squared Error (MSE). Minimizing this error leads us to the "line of best fit," which captures the underlying data trend in the simplest possible form.

3 Mean Squared Error (MSE)

The core concept of calculating MSE is to measure how much the model's prediction is accurate. We start with the basic error term $(\hat{y} - y)$, and by squaring and averaging over all data points, we obtain the Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

As mentioned the goal is to minimize the error between actual value and predicted value so we have the equation below as cost function:

$$J_{WB} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

When the value of this equation reaches its minimum, we can conclude that the model has found the best-fitting line for the data.

3.1 How to minimize the error?

There are two common approaches to minimizing error and enabling the model to learn from data. The first, a basic and classical method, is solving the optimization problem using matrix multiplication, which is effective for small datasets with a limited number of features. The second approach is Gradient Descent, a more scalable method that iteratively adjusts parameters and is well-suited for larger datasets and higher-dimensional problems.

4 Gradient Descent

The goal of gradient descent is to minimize the error (MSE) as much as possible. This approach can be accessed by two main formulas to calculate best fitted w and b

$$w := w - \alpha \frac{\partial J}{\partial w} \quad , \quad b := b - \alpha \frac{\partial J}{\partial b}$$

where α is the learning rate that controls the step size of the updates.

4.1 Compute the Gradients

To update the parameters, we first compute the partial derivatives of the cost function $J(w, b)$ with respect to w and b . These gradients measure how sensitive the error is to small changes in each parameter:

$$\begin{aligned} \frac{\partial J}{\partial w} &= \frac{2}{m} \sum_{i=1}^m \left((wx^{(i)} + b) - y^{(i)} \right) x^{(i)} \\ \frac{\partial J}{\partial b} &= \frac{2}{m} \sum_{i=1}^m \left((wx^{(i)} + b) - y^{(i)} \right) \end{aligned}$$

4.2 Update Intuition

- If the predictions are too high, the gradients are positive. Subtracting them decreases w and b , bringing predictions down. - If the predictions are too low, the gradients are negative. Subtracting them increases w and b , raising predictions.

In this way, the parameters iteratively “slide downhill” on the cost function surface toward the values that minimize the error.

5 Example

Let's consider a very simple dataset with just two points:

$$(x, y) = (1, 2), \quad (2, 4)$$

The true relationship is $y = 2x$.

Suppose we start with initial parameters:

$$w = 0, \quad b = 0$$

Predictions and errors:

$$\hat{y}(1) = 0 \Rightarrow \text{error} = -2$$

$$\hat{y}(2) = 0 \Rightarrow \text{error} = -4$$

Compute gradients:

$$\frac{\partial J}{\partial w} = \frac{2}{2} [(-2)(1) + (-4)(2)] = -10$$

$$\frac{\partial J}{\partial b} = \frac{2}{2} [(-2) + (-4)] = -6$$

Update step (with learning rate $\alpha = 0.1$):

$$w := 0 - 0.1(-10) = 1$$

$$b := 0 - 0.1(-6) = 0.6$$

Result:

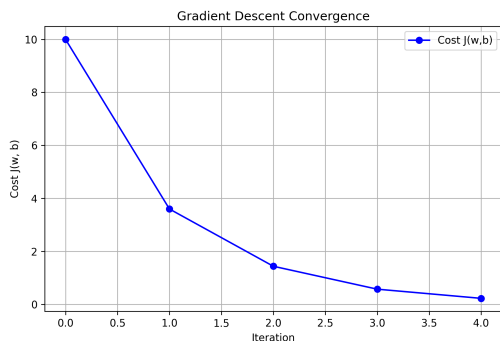
$$w = 1, \quad b = 0.6$$

Already, the parameters are closer to the true values $w = 2$, $b = 0$. Repeating this process iteratively will make them converge.

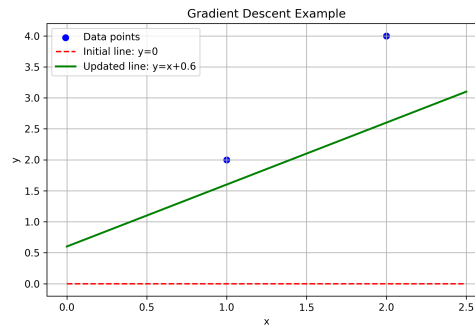
Gradient Descent iteratively updates the parameters w and b to minimize the cost function. Starting from $w = 0$ and $b = 0$, after each step, the parameters move closer to the optimal values ($w = 2$, $b = 0$).

For our tiny dataset, after a few iterations, the parameters converge:

Iteration	w	b
0	0	0
1	1	0.6
2	1.6	0.96
3	1.92	1.15
4	1.984	1.2



figureCost function decreasing over iterations.



figureData points and regression lines: initial line $y = 0$ and first updated line $y = x + 0.6$.

Thanks for coming so far, you can contact us on LinkedIn.
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